A Linear Maximum Torque Per Ampere Control for IPMSM Drives Over Full-Speed Range

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***Abstract*—In this paper, a linear torque control strategy is first proposed for interior permanent magnet synchronous motor drives to fully utilize the reluctance torque and simplify the controller design. The proposed linear torque control strategy also extends the existing maximum torque per ampere control in the constant torque limit region up to the entire field-weakening region. It is found that in an intermediate speed region, called partial field-weakening region, the existing maximum torque per ampere control can still be applied under lighter load condition. In addition, the proposed control can also achieve the objective of minimum copper loss (i.e., maximum torque per ampere) for the entire speed range. Sound theoretical basis is given in the context. Moreover, an adaptive limiter is proposed for efficiently implementing the proposed control strategy over the entire speed range. Finally, a prototype is also constructed by using a fixed-point DSP TMS320F240 and some experimental results are given to verify the validity of the proposed control strategy.**

***Index Terms*—Constant torque limit control, field-weakening control, interior permanent magnet synchronous motor (IPMSM), linear torque control.**

NOMENCLATURE

i_{ds}, i_{qs}	Stator d- and q-axis currents, respectively.
v_{ds}, v_{qs}	Stator d- and q-axis voltages, respectively.
$\mathrm{L_d}$, $\mathrm{L_q}$	d- and q-axis inductances, respectively.
$\mathrm{R_S}$	Stator winding resistance.
λ	Rotor permanent magnet flux linkage.
p	Number of poles.
T_e	Electromagnetic torque.
ω_r	Electrical angular frequency of the motor.
$\mathrm{V_{SM}}$	Maximum phase voltage amplitude.
$\mathrm{I_{SM}}$	Maximum line current amplitude.
I_{dsM}, I_{qsM}	Maximum d- and q-axis currents, respectively, for $\omega_r < \omega_{rM}$.
ω_{rM}	Boundary angular frequency corresponding to I_{dsM}, I_{qsM}, and $\mathrm{V_{SM}}$.
T_{eM}	Maximum torque for $\omega_r < \omega_{rM}$.
u_M	Maximum virtual control for $\omega_r < \omega_{rM}$.
ω_{rC}	Critical angular frequency.
ω_{rE}	Extreme angular frequency.
i_{dsb}, i_{qsb}	Boundary d- and q-axis currents, respectively.
T_{eb}	Electromagnetic torque bound.
u^*	Virtual control.
u_b	Virtual control bound.
i_{dsm}, i_{qsm}	Maximum d- and q-axis currents respectively for $\omega_r \in (\omega_{rM}, \omega_{rC})$.
T_{em}	Maximum torque for $\omega_r \in (\omega_{rM}, \omega_{rC})$.
u_m	Maximum virtual control for $\omega_r \in (\omega_{rM}, \omega_{rC})$.

Manuscript received November 6, 2003; revised March 13, 2004. Paper no. TEC-00325-2003.

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Digital Object Identifier 10.1109/TEC.2004.841517

I. INTRODUCTION

RECENTLY, the interior permanent magnet synchronous motor (IPMSM) drives are getting more popular in many applications such as traction and machine tool spindle drives [1], air conditioning compressors [2], [3], electrical vehicles [4], [5], and integrated starters/alternators [6]–[8]. The reason why an IPMSM has become so popular is mainly due to its many attractive characteristics such as high efficiency, high power density, high torque-to-inertia ratio, wide speed operation range, and free from maintenance.

Although there are many vector controls available in the existing literature for variable speed applications of IPMSMs, basically, they can be classified as either linear torque control strategy or nonlinear torque control strategy. For example, by letting the d-axis stator current (i_{ds}) be equal to zero such that the q-axis current (i_{qs}) is proportional to the torque command, it is quite straightforward to achieve a linear torque control [9]–[11] to achieve high performance. However, the potential reluctance torque of the IPMSM is not fully employed. On the other hand, to take advantage of the reluctance torque, many nonlinear torque control strategies such as unity power factor control [12], constant flux linkage control [12], maximum torque per ampere (MTPA) control [13]–[17], and maximum efficiency control [18], [19] were developed to achieve different objectives. For the above mentioned nonlinear controls, the corresponding q-axis current command was first generated by the demanded torque and the corresponding i_{ds} was then decided according to the specific objective. However, this renders the design of the resulting controller more difficult. In view of the above problems, it is the major motivation of this paper to propose a linear maximum torque per ampere control strategy to achieve fast transient response and, meanwhile, simplify the corresponding controller design. In fact, this linear control strategy is especially convenient for the synchronous control of multiple IPMSM drives.

The remaining contents of this paper are organized as follows. In Section II, the proposed linear maximum torque per ampere control is first presented. Then, the extension to the field-weakening region is followed in Section III. Also an implementation and some experimental results of the proposed strategy are provided in Section IV. Finally, some conclusions are offered in the last section.

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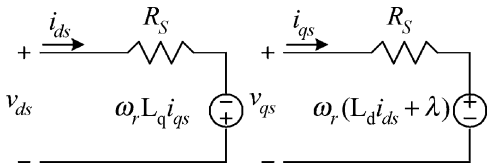


Fig. 1. d- and q-axis steady state equivalent circuits in the rotor reference frame of an IPMSM.

II. Proposed Linear Maximum Torque per Ampere Control—Constant Torque Limit Region

For reference, the steady-state d- and q-axis equivalent circuits of an IPMSM in the rotor reference frame are given in Fig. 1 [20]. The corresponding d- and q-axis voltage equations can be obtained from Fig. 1 as follows:

$$v_{ds} = \mathrm{R_S} i_{ds} - \omega_r \mathrm{L_q} i_{qs} \tag{1}$$

$$v_{qs} = \mathrm{R_S} i_{qs} + \omega_r(\mathrm{L_d} i_{ds} + \lambda). \tag{2}$$

The corresponding generated electromagnetic torque is given as

$$T_\mathrm{e} = 0.75p\lambda i_{qs} + 0.75p(\mathrm{L_d} - \mathrm{L_q}) i_{ds} i_{qs} \tag{3}$$

where p is the number of poles. Also, for a practical inverter-fed motor drive, there exists an inverter output voltage limit and an output current limit. Assume that the maximum available phase voltage amplitude and the maximum line current amplitude are $\mathrm{V_{SM}}$ and $\mathrm{I_{SM}}$, respectively. Then the feasible operation range is constrained by the following inequalities:

$$i_{ds}^2 + i_{qs}^2 \le \mathrm{I_{SM}^2} \tag{4}$$

$$v_{ds}^2 + v_{qs}^2 \le \mathrm{V_{SM}^2}. \tag{5}$$

From (3), one can observe that when i_ds is equal to zero, then the resulting T_e is proportional to i_{qs}. Hence, it is quite simple to implement a high-performance drive as in [9]–[11]. However, the reluctance torque of the IPMSM is not fully exploited. In the case where i_ds is not equal to zero, then it is seen that T_e contains a nonlinear term, namely $0.75\mathrm{p}(\mathrm{L_d} - \mathrm{L_q}) i_{ds} i_{qs}$ rendering the controller design much more difficult than a linear one.

As far as achieving fast transient response is concerned, the maximum torque per ampere control [13], [14] is rather attractive. For a given torque demand, the line current amplitude, or $\sqrt{i_{ds}^2 + i_{qs}^2}$ equivalently, is minimized to achieve the maximum torque. Within the current constraint of (4), the d- and q-axis stator currents should satisfy the following equation [14]:

$$f(i_{ds}, i_{qs}) = 2(\mathrm{L_q} - \mathrm{L_d}) i_{ds} - \lambda + \sqrt{\lambda^2 + 4(\mathrm{L_q} - \mathrm{L_d})^2 i_{qs}^2} = 0. \tag{6}$$

In particular, when the maximum line current amplitude $\mathrm{I_{SM}}$ is imposed, namely

$$i_{ds}^2 + i_{qs}^2 = \mathrm{I_{SM}^2} \tag{7}$$

then from (6) and (7), one can obtain the maximum available torque T_{eM} as follows:

$$T_{eM} = 0.75\mathrm{p}\left[\lambda + (\mathrm{L_d} - \mathrm{L_q}) I_{dsM}\right] I_{qsM} \tag{8}$$

where

$$I_{dsM} \triangleq \frac{\lambda - \sqrt{\lambda^2 + 8(\mathrm{L_q} - \mathrm{L_d})^2 \mathrm{I_{SM}^2}}}{4(\mathrm{L_q} - \mathrm{L_d})} \tag{9}$$

$$I_{qsM} \triangleq \sqrt{\mathrm{I_{SM}^2} - I_{dsM}^2}. \tag{10}$$

Also, the corresponding angular speed, namely ω_{rM} can be calculated by substituting (1), (2) and (9), (10) into the following voltage boundary equation:

$$(\mathrm{R_s} I_{dsM} - \omega_{rM} \mathrm{L_q} I_{qsM})^2 + \left[\mathrm{R_s} I_{qsM} + \omega_{rM}(\mathrm{L_d} I_{dsM} + \lambda)\right]^2 = \mathrm{V_{SM}^2}. \tag{11}$$

Thus, whenever there exists a speed deviation during the control process, the maximum torque T_{eM} can be applied to achieve the fastest response. Next, in order to achieve a linear control law, a virtual control u^*, corresponding to a torque command T_e^*, can be defined as follows:

$$u^* \triangleq \frac{T_e^*}{\mathrm{K_{tM}}} = \frac{0.75\mathrm{p}\left[\lambda + (\mathrm{L_d} - \mathrm{L_q}) i_{ds}^*\right] i_{qs}^*}{\mathrm{K_{tM}}} \tag{12}$$

where constant $\mathrm{K_{tM}}$ is chosen to be

$$\mathrm{K_{tM}} \triangleq \frac{T_{eM}}{\mathrm{I_{SM}}}. \tag{13}$$

As will be clear from later sections, the same constant, $\mathrm{K_{tM}}$, is used not only in the constant torque limit range but also in the entire field-weakening range. In fact, the discovery of the proposition of the constant $\mathrm{K_{tM}}$ is inspired by the authors from another linear torque control strategy [21] of IPMSM. For convenient explanation of the proposed linear maximum torque per ampere (LMTPA) control, define Region I as the constant torque limit region where $\omega_r < \omega_{rM}$. Hence, in this region, if a torque command $T_e^* (= \mathrm{K_{tM}} u^*)$ is given, then the corresponding d-axis current command (i_{ds}^*) can be solved from the following polynomial equation:

$$\begin{aligned} P_1(i_{ds}^*) \triangleq\ & (\mathrm{L_d} - \mathrm{L_q})^3 i_{ds}^{*4} + 3(\mathrm{L_d} - \mathrm{L_q})^2 \lambda i_{ds}^{*3} \\ & + 3(\mathrm{L_d} - \mathrm{L_q}) \lambda^2 i_{ds}^{*2} + \lambda^3 i_{ds}^* - \frac{16}{9\mathrm{p}^2} (\mathrm{L_d} - \mathrm{L_q}) \mathrm{K_{tM}^2} u^{*2} \\ =\ & 0 \end{aligned} \tag{14}$$

which is obtained by substituting

$$i_{qs} = \frac{T_e^*}{0.75\mathrm{p}\left[\lambda + (\mathrm{L_d} - \mathrm{L_q}) i_{ds}^*\right]} = \frac{\mathrm{K_{tM}} u^*}{0.75\mathrm{p}\left[\lambda + (\mathrm{L_d} - \mathrm{L_q}) i_{ds}^*\right]} \tag{15}$$

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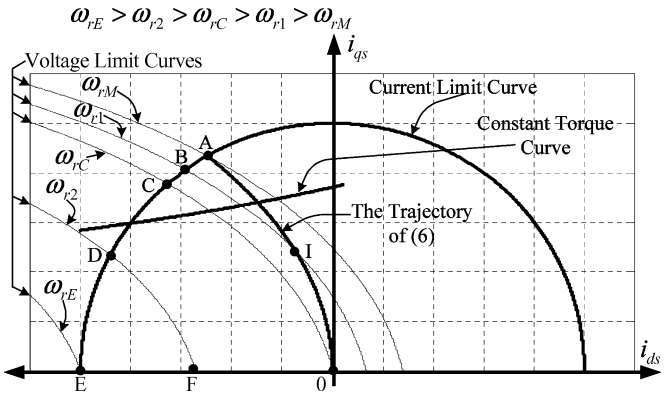


Fig. 2. Trajectories of (6), the current limit curve, one constant torque curve, and five voltage limit curves on $i_{ds} - i_{qs}$ plane for an IPMSM.

into (6). Similarly, by substituting

$$i_{ds} = \frac{\left(\frac{\mathrm{K_{tM}}u^*}{0.75\mathrm{p}} - \lambda i_{qs}^*\right)}{\left((\mathrm{L_d} - \mathrm{L_q})i_{qs}^*\right)} \tag{16}$$

into (6), one can obtain the following relation:

$$P_2\left(i_{qs}^*\right) \triangleq 9\mathrm{p}^2(\mathrm{L_d}-\mathrm{L_q})^2 i_{qs}^{*4} + 12\mathrm{p}\mathrm{K_{tM}}\lambda u^* i_{qs}^* - 16\mathrm{K_{tM}^2}u^{*2} = 0. \tag{17}$$

The corresponding q-axis current command i_{qs}^* can be solved from (17). Although the orders of (14) and (17) look rather high, as will be clear in a later implementation example, a second-order polynomial approximation is, in fact, accurate enough. The desired i_{ds}^* and i_{qs}^* can be obtained easily by using any available commercial numerical program with zero initial values. For clarity, the trajectory of (6) on the $i_{ds} - i_{qs}$ plane, namely, the AIO curve, is shown in Fig. 2, where the current limit curve and one constant torque curve together with five voltage limit curves are also shown on the same figure. In summary, in Region I, given a torque command T_e^*, one can get the corresponding virtual control u^*. Then, from u^*, one can obtain the corresponding i_{ds}^* and i_{qs}^* from (14) and (17), respectively. Although the generated torque T_e^* as in (3) is still nonlinear to i_{ds}^* and i_{qs}^*, it is now proportional to the virtual control u^* as in (12). Thus, excluding the inner current-loop control, the outer-loop controller can be easily designed by using any existing linear control strategy directly. Since the AIO curve in Fig. 2 is identical with the maximum torque per ampere control [14], the proposed linear control can achieve the same dynamic performance as in [14].

III. Extension of the Proposed Control to the Field-Weakening Region

From the previous section, it is seen that during acceleration or deceleration, for $\omega_r < \omega_{rM}$, one can choose the maximum torque T_{eM} to achieve the fastest response. However, when $\omega_r > \omega_{rM}$, due to the current and voltage constraints of (4) and (5), the maximum torque (T_{eM}) cannot be achieved. The traditional control strategy is simply to reduce the magnetic field intensity by applying negative i_{ds} resulting in the so called field-weakening control. As can be observed from Fig. 2, as far as the current constraint is concerned, the field-weakening region is bounded by the ABCDE curve, AIO curve, and the EO axis. In particular, four points (namely B, C, D, E) are also marked in Fig. 2, where the voltage limit is also imposed on, namely

$$v_{ds}^2 + v_{qs}^2 = \mathrm{V_{SM}^2}. \tag{18}$$

Substituting (1) and (2) into (18) yields

$$\begin{aligned} \mathrm{g}(i_{ds}, i_{qs}, \omega_r) \triangleq\ & (\mathrm{R_S}i_{ds} - \omega_r \mathrm{L_q} i_{qs})^2 \\ & + (\mathrm{R_S}i_{qs} + \omega_r \mathrm{L_d} i_{ds} + \omega_r \lambda)^2 - \mathrm{V_{SM}^2} \\ & = 0. \end{aligned} \tag{19}$$

Thus, from (19), it is seen that when $i_{ds} = i_{qs} = 0$, then one can obtain the corresponding speed, called the critical angular frequency ω_{rC}

$$\omega_{rC} = \frac{\mathrm{V_{SM}}}{\lambda}. \tag{20}$$

Similarly, from (19), when $i_{ds} = -\mathrm{I_{SM}}$ and $i_{qs} = 0$ then one can obtain the corresponding speed, called the extreme angular frequency ω_{rE}

$$\omega_{rE} = \frac{\sqrt{\mathrm{V_{SM}^2} - (\mathrm{R_s I_{SM}})^2}}{\lambda - \mathrm{L_d I_{SM}}}. \tag{21}$$

The corresponding operating points of ω_{rC} and ω_{rE} on the $i_{ds} - i_{qs}$ plane are just the C, E points marked in Fig. 2. Also, from (21), one can see that to get a positive value of ω_{rE}, it is necessary that λ is greater than $\mathrm{L_d I_{SM}}$. Indeed, in practical operation situations, $\mathrm{L_d}|i_{ds}|$ is less than λ. Further examination of Fig. 2 reveals that when the motor speed $\omega_r \in (\omega_{rM}, \omega_{rC})$, then the voltage bound curve of (19) will intersect with the AIO curve. For example, Fig. 2 shows a voltage bound curve for $\omega_r = \omega_{r1}$, and the corresponding intersecting point is I. In other words, for each given $\omega_r \in (\omega_{rM}, \omega_{rC})$, the corresponding boundary currents i_{dsb} and i_{qsb} of point I can be solved from the following system of polynomials, namely (6) and (19)

$$f(i_{dsb}, i_{qsb}) = 0 \tag{22a}$$

$$g(i_{dsb}, i_{qsb}, \omega_r) = 0. \tag{22b}$$

Hence, the corresponding electromagnetic torque bound and virtual control bound, namely T_{eb} and u_b, can be defined as follows:

$$T_{eb} \triangleq 0.75\mathrm{p}\left[\lambda + (\mathrm{L_d} - \mathrm{L_q})i_{dsb}\right] i_{qsb} \tag{23a}$$

$$u_b \triangleq \frac{T_{eb}}{\mathrm{K_{tM}}}. \tag{23b}$$

Now it is quite clear from Fig. 2 that when $\omega_r = \omega_{r1}$ and $T_e^* < T_{eb}$, then despite that ω_{r1} is located in the conventional field-weakening region, one can still use the maximum torque per ampere control, namely corresponding to the curve IO portion of Fig. 2, to achieve the minimum copper loss. Also, the corresponding u^*, as well as i_{ds}^* and i_{qs}^*, remains the same as

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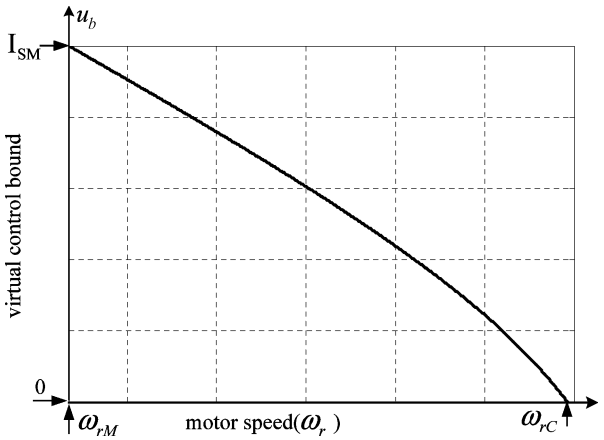


Fig. 3. Graphical illustration of the virtual control bound to motor speed curve for $\omega_{rM} \leq \omega_r \leq \omega_{rC}$.

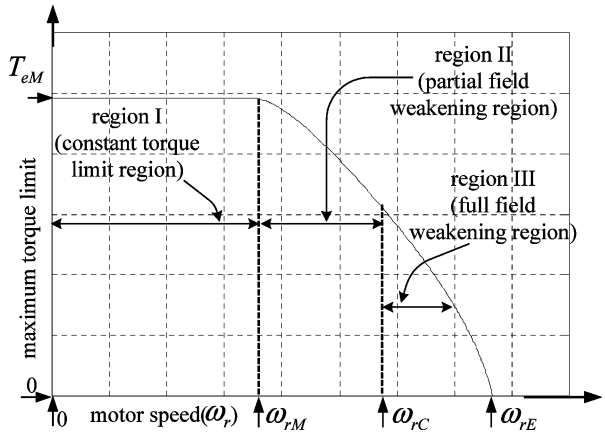


Fig. 4. Illustration of the maximum torque limit to motor speed curve and three operation regions for an IPMSM.

that obtained from (14) and (17). However, when $T_e^* > T_{eb}$, due to the voltage constraint, it is not possible to achieve the conventional maximum torque per ampere control. Therefore, the proposed control strategy chooses i_{ds}^* and i_{qs}^*, for a given T_e^* (and $u^* = (T_e^*/\mathrm{K_{tM}})$), as the intersection point of the torque command equation (3) and the voltage limit equation (19). Similar to the previous constant torque limit control, for convenience, by using (3), one can either eliminate i_{qs}^* to yield

$$\begin{aligned} P_{1w}\left(i_{ds}^*\right) = &\left[R_s i_{ds}^* - \omega_r \mathrm{L_q} \frac{\mathrm{K_{tM}} u^*}{0.75\mathrm{p}\left[\lambda + (\mathrm{L_d} - \mathrm{L_q}) i_{ds}^*\right]}\right]^2 \\ &+ \left[\frac{R_s \mathrm{K_{tM}} u^*}{0.75\mathrm{p}\left[\lambda + (\mathrm{L_d} - \mathrm{L_q}) i_{ds}^*\right]} + \omega_r (\mathrm{L_d} i_{ds}^* + \lambda)\right]^2 \\ &- \mathrm{V_{SM}^2} \\ &= 0 \end{aligned} \tag{24}$$

or eliminate i_{ds}^* to yield

$$\begin{aligned} P_{2w}\left(i_{qs}^*\right) = &\left[\frac{\mathrm{R_s}\left(\frac{\mathrm{K_{tM}} u^*}{0.75\mathrm{p}} - \lambda i_{qs}^*\right)}{\left((\mathrm{L_d} - \mathrm{L_q}) i_{qs}^*\right)} - \omega_r \mathrm{L_q} i_{qs}^*\right]^2 \\ &+ \left[\mathrm{R_s} i_{qs}^* + \omega_r \left(\frac{\mathrm{L_d}\left(\frac{\mathrm{K_{tM}} u^*}{0.75\mathrm{p}} - \lambda i_{qs}^*\right)}{\left((\mathrm{L_d} - \mathrm{L_q}) i_{qs}^*\right)} + \lambda\right)\right]^2 \\ &- \mathrm{V_{SM}^2} \\ &= 0. \end{aligned} \tag{25}$$

In summary, for $\omega_r \in (\omega_{rM}, \omega_{rC})$, if $u^* < u_b$ (or equivalently $T_e^* < T_{eb}$), then the previous maximum torque per ampere control can be applied. If $u^* \geq u_b$, then the field-weakening control which is obtained from (24) and (25) should be applied. Since there are two control modes applicable in this speed range, it is called the partial field-weakening region (Region II). It is worth mentioning that within the partial field-weakening region, the virtual control bound u_b is not a constant. It is varied with motor speed ω_r. For reference, Fig. 3 also shows the trajectory of a virtual control bound within the partial field-weakening region as an illustration.

Furthermore, for $\omega_r \in (\omega_{rC}, \omega_{rE})$, it is seen from Fig. 2 that it is not possible to have an intersection point for (6) and (19). Thus, for each torque command T_e^* (or equivalently u^*), the same field-weakening control equations (24) and (25) are directly applied to find the corresponding i_{ds}^* and i_{qs}^*. Therefore, in this speed region, it is called the full field-weakening region (Region III). For reference, Fig. 4 shows a schematic diagram for the maximum available torque with respect to the whole speed range. In fact, one of the characteristics of the proposed control strategy is that under transient condition, no matter the operation speed is located in which region, the maximum torque capability can always be applied to achieve the fastest response. From previous results, the proposed control strategy has been shown to be a linear one over the entire speed region. Also, for Region I and a portion of Region II, the proposed control is identical with the maximum torque per ampere control to achieve minimum copper loss.

It remains to be proven that the proposed field-weakening control can also achieve the minimum copper loss. By using (3), one can get

$$\begin{aligned} i_{ds}^2 + i_{qs}^2 &\triangleq h(i_{ds}) \\ &= i_{ds}^2 + \left(\frac{T_e}{0.75\mathrm{p}\left[\lambda + (\mathrm{L_d} - \mathrm{L_q}) i_{ds}\right]}\right)^2. \end{aligned} \tag{26}$$

It follows from (26) that:

$$\frac{dh(i_{ds})}{di_{ds}} = 2i_{ds} - \frac{1.5\mathrm{p} T_e^2 (\mathrm{L_d} - \mathrm{L_q})}{\left[0.75\mathrm{p}\left(\lambda + (\mathrm{L_d} - \mathrm{L_q}) i_{ds}\right)\right]^3}. \tag{27}$$

Equation (27) can be further simplified by substituting (3) to obtain

$$\frac{dh(i_{ds})}{di_{ds}} = 2i_{ds} - \frac{2i_{qs}^2 (\mathrm{L_d} - \mathrm{L_q})}{\lambda + (\mathrm{L_d} - \mathrm{L_q}) i_{ds}}. \tag{28}$$

It follows from (28) that in case the derivative is set equal to zero, one can obtain the familiar $i_{ds} \sim i_{qs}$ relation for the maximum torque per ampere control [14] as follows:

$$i_{ds} = \frac{\lambda - \sqrt{\lambda^2 + 4(\mathrm{L_d} - \mathrm{L_q})^2 i_{qs}^2}}{2(\mathrm{L_q} - \mathrm{L_d})}. \tag{29}$$

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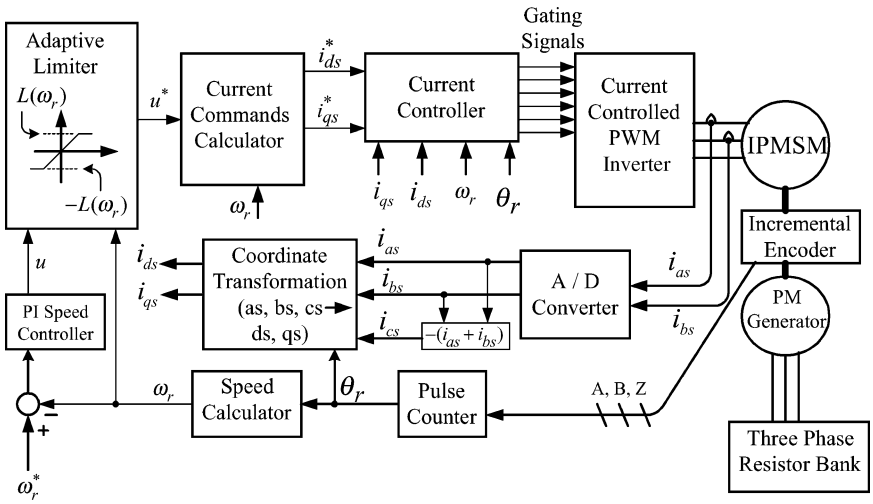


Fig. 5. Block diagram of the developed IPMSM drive system.

Thus, when the proposed field-weakening control is applied in either Region III or Region II for $T_e^* > T_{eb}$, it can be observed from Fig. 2 that the corresponding i_{ds} will be less than that of (29), namely

$$i_{ds} < \frac{\lambda - \sqrt{\lambda^2 + 4(\mathrm{L_d} - \mathrm{L_q})^2 i_{qs}^2}}{2(\mathrm{L_q} - \mathrm{L_d})}. \tag{30}$$

It follows from (28) and (30) that:

$$\frac{dh(i_{ds})}{di_{ds}} < 0. \tag{31}$$

In other words, $h(i_{ds}) = i_{ds}^2 + i_{qs}^2$ is a monotonically decreasing function with increasing i_{ds}. Furthermore, from (3) one has

$$i_{qs} = \frac{T_\mathrm{e}}{0.75\mathrm{p}\left[\lambda + (\mathrm{L_d} - \mathrm{L_q}) i_{ds}\right]}. \tag{32}$$

Hence, for a fixed positive torque T_e, one can obtain the following derivative:

$$\frac{di_{qs}}{di_{ds}} = \frac{0.75\mathrm{p}T_\mathrm{e}(\mathrm{L_q} - \mathrm{L_d})}{\left[0.75\mathrm{p}\left(\lambda + (\mathrm{L_d} - \mathrm{L_q}) i_{ds}\right)\right]^2}. \tag{33}$$

It is seen from (33) that the slope at a point located on the constant torque contour is positive as is illustrated by the constant torque trajectory shown in Fig. 2. On the other hand, it can be observed from Fig. 2 that under the proposed field-weakening control, the corresponding i_{ds}^* is determined by the voltage limit equation (19) and the constant torque (T_e^*) trajectory, implying that the i_{ds}^* is the largest one within the feasible constant torque trajectory. Hence, the proposed linear control can also achieve the minimum copper loss in the field-weakening region.

Finally, consider the transient control. For example, during the motor starting stage, since $\omega_r < \omega_{rM}$, the maximum torque (T_{eM}) can be applied to achieve the fastest response. However, when the motor speed is greater than ω_{rM}, the maximum torque T_{eM} cannot be achieved. For the proposed LMTPA control and for each ω_r, the corresponding maximum torque, called T_{em}, can be obtained by first finding the corresponding d- and q-axis currents, namely I_{dsm} and I_{qsm} from (7) and (19)

$$I_{dsm}^2 + I_{qsm}^2 = \mathrm{I_{SM}^2} \tag{34a}$$

$$g(I_{dsm}, I_{qsm}, \omega_r) = 0. \tag{34b}$$

Then, the maximum torque and maximum virtual control becomes

$$T_{em}(\omega_r) = 0.75\mathrm{p}\left[\lambda + (\mathrm{L_d} - \mathrm{L_q}) I_{dsm}\right] I_{qsm} \tag{35a}$$

$$u_m(\omega_r) = \frac{T_{em}(\omega_r)}{\mathrm{K_{tM}}}. \tag{35b}$$

It should be mentioned that both I_{dsm} and I_{qsm} are functions of ω_r for $\omega_r \in (\omega_{rM}, \omega_{rE})$.

IV. Implementation and Experimental Results of the Proposed LMTPA Control

According to previous LMTPA control, an implementation block diagram of the speed control is shown in Fig. 5 as an illustrative example. It is seen from Fig. 5 that ω_r^* is a speed command. The signals of i_{as} and i_{bs} are the line currents of phases a and b, respectively. The pulse signals of A, B, and Z are the outputs of the incremental encoder, where A and B are two sequences of pulses with variable frequencies and fixed phase shifts of 90 degrees. The Z signal is used for resetting when the rotor of the motor revolves one cycle, a signal pulse will appear in Z. Except for the adaptive limiter block and the current command calculator, the other blocks are quite typical and will not be explained further. First, consider the adaptive limiter block. From the proposed control strategy, it is seen that for $\omega_r \le \omega_{rM}$, the maximum virtual control available is chosen to be

$$u_M = \frac{T_{eM}}{\mathrm{K_{tM}}} = \mathrm{I_{SM}}, \quad \omega_r \le \omega_{rM}. \tag{36}$$

Also, for $\omega_r \in (\omega_{rM}, \omega_{rE})$, the maximum virtual control available is $u_m(\omega_r)$ from (35b). For easy implementation, one can

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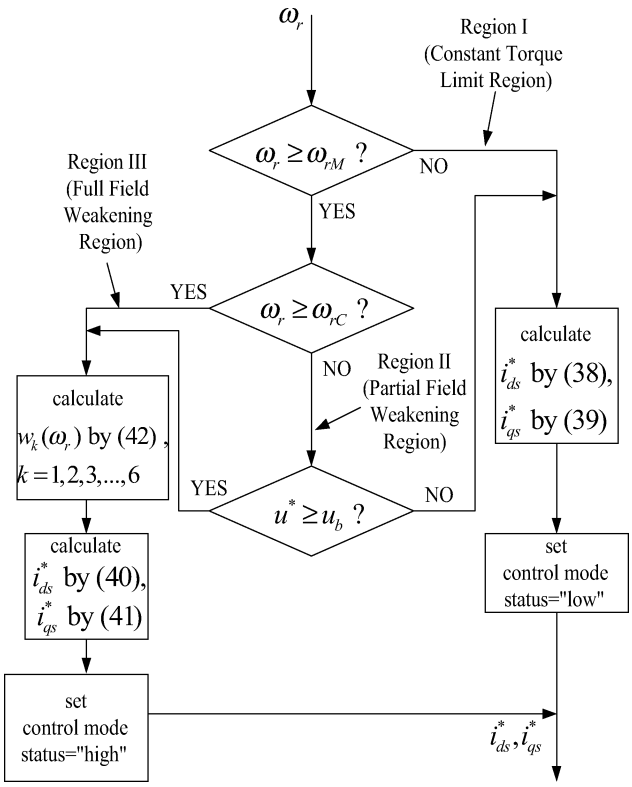


Fig. 6. Flowchart of the current commands calculator as reference to Fig. 5.

define an adaptive limiter with the following adaptive upper bound:

$$L(\omega_r) \triangleq \begin{cases} u_M = \mathrm{I_{SM}}, & 0 \leq \omega_r \leq \omega_{rM} \\ u_m(\omega_r), & \omega_{rM} < \omega_r < \omega_{rE} \end{cases}. \quad (37)$$

The shape of the adaptive upper bound $L(\omega_r)$ over the entire speed range is the same as in Fig. 4 except that the unit of the vertical axis is ampere. Of course, when the output of the speed controller is not saturated, then the corresponding virtual control u^* is the same as the original u in this linear control region as shown in Fig. 5. Next, consider the current command calculator. Fig. 6 shows the detailed flowchart of how to calculate the desired i_{ds}^* and i_{qs}^* for each virtual control u^* in different speed regions according to the proposed control strategy. However, for simplifying the complicated calculation, the solutions of (14) and (17) are calculated offline over the entire range of u^*. Then the equations i_{ds}^* and i_{qs}^* can be obtained by using the second order least square error approximation method. These two equations are listed as follows for the constant torque limit control:

$$i_{ds}^* = m_1(u^*)^2 + m_2 u^* + m_3 \quad (38)$$
$$i_{qs}^* = m_4(u^*)^2 + m_5 u^* + m_6 \quad (39)$$

where m_1 to m_6 are real constants. Similarly, for each $\omega_r \in (\omega_{rM}, \omega_{rE})$ in the field-weakening control, the solutions of (24) and (25) can be approximated by the following expressions:

$$i_{ds}^* = w_1(u^*)^2 + w_2 u^* + w_3 \quad (40)$$
$$i_{qs}^* = w_4(u^*)^2 + w_5 u^* + w_6. \quad (41)$$

TABLE I
TESTED IPMSM PARAMETERS AND INVERTER RATINGS

parameters \ tested IPMSM	
number of poles	6
armature resistance (Ω)	0.15
d-axis inductance (mH)	0.488
q-axis inductance (mH)	1.01
magnetic flux linkage (V/(rad/sec))	0.042
maximum phase voltage amplitude V_{SM} (V)	50
maximum phase current amplitude I_{SM} (A)	25.0

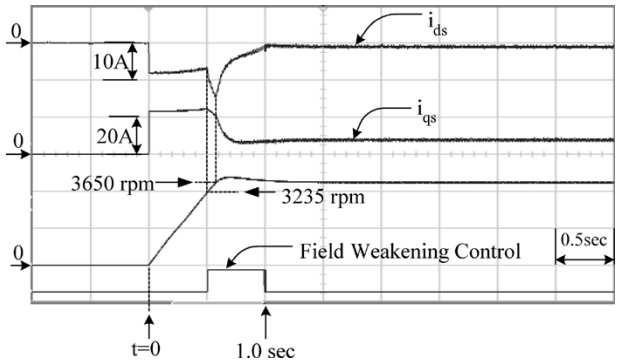


Fig. 7. Experimental results of i_{ds}, i_{qs}, and motor speed responses for the proposed control in the partial field-weakening region under lighter load.

Since the values of w_1 to w_6 in (40) and (41) now varies with ω_r for $\omega_r \in (\omega_{rM}, \omega_{rE})$, the second order least square error approximation method can also be used to find the equations of w_k, $k = 1, 2, \ldots, 6$ as

$$w_k = c_{k1}\omega_r^2 + c_{k2}\omega_r + c_{k3} \quad (42)$$

with c_{kj}, $k = 1, 2, \ldots, 6$, $j = 1, 2, 3$ being real constants. From actual experiments, it is found that, with the above approximations, the errors between the actual current commands and the approximated current commands for both d- and q-axis are less than 1%, but the calculation time is greatly reduced.

In order to verify the feasibility of the proposed control strategy, a prototype is constructed according to the block diagram of Fig. 5 by using a fixed-point DSP TMS320F240. The sampling time periods for the current controller and the speed controller are chosen to be 0.1 and 1 ms, respectively. The tested IPMSM, a prototype designed by the authors' laboratory, has the parameters as shown in Table I.

The corresponding ω_{rM} and ω_{rC} are found to be 3235 and 3790 r/min from (11) and (20), respectively. Similarly, the I_{dsM} and I_{qsM} are -6.6(A) and 24.1(A) from (9) and (10), respectively.

Due to limitation of space, only three test results are presented to illustrate the characteristics of the developed drive

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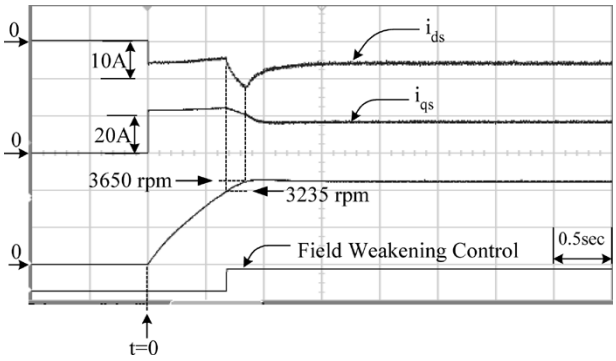


Fig. 8. Experimental results of i_{ds}, i_{qs}, and motor speed responses for the proposed control in the partial field-weakening region under heavier load.

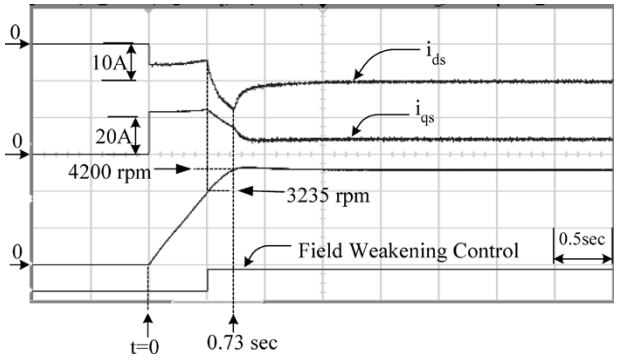


Fig. 9. Experimental results of i_{ds}, i_{qs}, and motor speed responses for the proposed control in the full field-weakening region.

system. To demonstrate the smooth transitions between the constant torque limit control mode and the field-weakening control mode in the partial field-weakening region, first consider the case of applying a step command of $\omega_r^* = 3650$ r/min to the tested IPMSM drive starting from rest at $\mathrm{t} = 0$ s and with lighter load such that the control is settled down to a constant torque limit control. Fig. 7 shows the d- and q-axis current responses, speed response, and the control mode status to show the transition between the constant torque limit control mode and the field-weakening control mode. As can be observed from Fig. 7, when the motor is accelerating up to ω_{rM} (= 3235 r/min), the motor is operated in Region I and the responses of i_{ds} and i_{qs} indeed follow approximately i_{dsM} and i_{qsM}, respectively. Also, the corresponding control mode status in Fig. 7 is at the low level. As soon as ω_r is greater than ω_{rM}, the motor is then operated in Region II. The responses of i_{ds} and i_{qs} decrease with the increase of ω_r until the motor speed ω_r reaches the speed command $\omega_r^* = 3650$ r/min. On the other hand, the control mode status in Fig. 7 has changed to the high level (the field-weakening control mode) at $\omega_r = 3235$ r/min. This field-weakening control mode persists until $\mathrm{t} = 1.0$ s, then the motor speed reaches the steady state speed $\omega_r = 3650$ r/min and the resulting control becomes the MTPA control. Meanwhile, the control mode status shows a second transition from the high level to the low level. It can be checked that the motor load torque T_L (approximately 0.9 Nm at 3650 r/min) is less than $\mathrm{K_{tM}}u_b$ (=1.7 Nm) at $\omega_r = 3650$ r/min. The second test condition is almost the same as the first one except that a heavier load is applied such that the final control is settled down to the field-weakening control. Fig. 8 shows the d- and q-axis current responses, speed response, and the corresponding control mode status for the same step speed command as Fig. 7. One can see that the control mode status in Fig. 8 shows only one transition from the low level to the high level (i.e., remaining in the field-weakening control mode). It can also be checked that this load torque T_L (approximately 3.0 Nm at 3650 r/min) is greater than the corresponding $\mathrm{K_{tM}}u_b$ (=1.7 Nm) at $\omega_r = 3650$ r/min.

Finally, to demonstrate the operation mode in Region III, a step command of $\omega_r^* = 4200$ r/min which is greater than $\omega_{rC} = 3790$ r/min is applied to the IPMSM drive starting from rest at $\mathrm{t} = 0$ s. Fig. 9 shows the corresponding d- and q-axis current responses, speed response, and the control mode status. As shown in Fig. 9, when the motor speed reaches $\omega_r = \omega_{rM} =$ 3235 r/min the control mode status also indicates a transition from low level to high level. When $\omega_r > \omega_{rM}$, then I_{dsm} and I_{qsm} both start decreasing until the motor speed ω_r is greater than the speed command $\omega_r^* = 4200$ r/min at about $\mathrm{t} = 0.73$ s. For $\mathrm{t} \geq 0.73$ s, the magnitudes of i_{ds} and i_{qs} gradually decrease to the steady-state values. However, since the motor speed is greater than ω_{rC} (= 3790 r/min), the steady state operation mode is still kept in the field-weakening control mode as can be observed from the control mode status of Fig. 9. For reference, Fig. 10 also shows the corresponding trajectory on the $i_{ds} - i_{qs}$ plane corresponding to the time responses of Fig. 9. It is seen that the current vector starts from zero to point A quickly and then the current trajectory moves approximately along the current limit curve until point D, where the motor speed reaches the command speed. Finally, the magnitude of the current vector decreases and then settles down to the steady state operating point to match the corresponding load torque. The above experimental tests show that the proposed LMTPA control can indeed achieve a linear control over the entire speed operation range and also the corresponding phenomena indeed agree with the theoretical prediction.

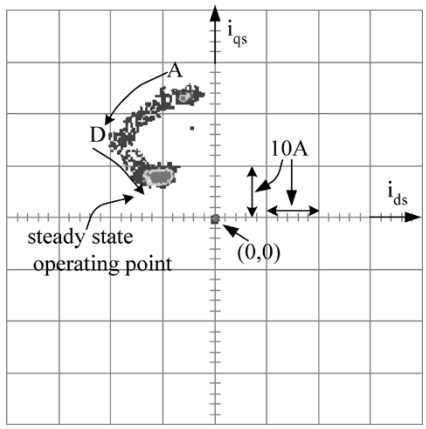


Fig. 10. $i_{ds} - i_{qs}$ plot corresponding to Fig. 9.

V. Conclusions

In this paper, a linear torque control strategy has been proposed by using a virtual control for the IPMSM drives to both fully utilize the reluctance torque and simplify the controller design. Also, it has successfully extended the existing maximum

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torque per ampere control in the constant torque limit region up to the field-weakening region. The theoretical basis of the proposed control strategy has been described in detail and some analytical forms are also provided for reference. Basically, the operation range of the drive system can be divided into three regions. The first one is called the constant torque limit region and Regions II and III correspond to the partial field-weakening region and the full field-weakening region respectively. However, only two control modes are required. One is the constant torque limit control, which can be applied to either Region I or part of Region II, and the other is the field-weakening control, which can be applied to either part of Regions II or III. Furthermore, within the operable region, the steady-state current vector command is proven to have minimum amplitude in both the constant torque limit control mode and the field-weakening control mode. Therefore, the proposed control can achieve minimum copper loss over the entire speed range. Moreover, for easy implementation, an adaptive limiter is proposed to fully utilize the maximum torque capability to achieve fast response. The proposed strategy has been successfully implemented fully digitally by using a fixed-point DSP TMS320F240 to simplify the hardware circuits. Finally, some experimental results are given to demonstrate the feasibility of the proposed control strategy.

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